The electron as a photon on a double loop

Williamson & van der Mark's toroidal-photon electron on the test bench, and the electron pumps circuits can build — register entry H23, CLAUDE.md §3i, §3j

kronspice simulation report · 2026-11-07 · magnetron GA seed 2450, 40 × 30

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Summary

Williamson and van der Mark (1997) proposed that the electron is one photon of the Compton wavelength going twice round a torus. This example puts the paper's numbers beside measurement, builds its 720° double loop as a circuit (a two-wire ring line with its conductors crossed once), and tests the paper's central assumption — that the light stays in the loop — with the field solver *em/fdtd.go*. It then builds three electron pumps that real circuits can make, each with an energy ledger, and evolves a multipactor pump driven by a microwave-oven magnetron.

- **The model's charge is 9.0 % low:** q = 0.910318 e, so its magnetic anomaly (g − 2)/2 = 0.000962436 is 17.0 % below the measured 0.00115965218.
- **The 720° property is reproduced as a circuit:** the crossed ring line has half-integer modes (54,548 Hz = ½ wavelength round the ring) and a pulse returns inverted after one trip (−0.91 of launch) and upright after two (+0.70).
- **Light does not stay in the loop:** under linear Maxwell equations the energy within 2R falls from 98.93 % to 5.773 % after a quarter period, and 99.99 % has left the region after 2 periods (ledger residual -1.9×10^{-13}). The paper assumes a confinement it does not supply.
- **Single-electron pump:** 1.000000000 electrons per cycle at 1 MHz (I = e·f = 0.1602 pA), the gates paying the heat.
- **Resonant ground pump:** 10 V at 100 kHz → 199.9 V peak → bank −1194.7 V against the ideal −1199.5 V; the earth lends 7.46×10^{14} electrons and gets the same number back.
- **Magnetron-driven multipactor pump, evolved:** 7.475 µA → 659.4 µA collected (176.6 → 2.002 mains W per µA). Arrays of gaps reach 1.95 A (LG 2M214, 1.32 kW mains) and 216 A (Stellant CWM-100kW, 119 kW mains).

What is being tested

Hypothesis (H23). J. G. Williamson and M. B. van der Mark, “Is the electron a photon with toroidal topology?”, *Ann. Fond. Louis de Broglie* 22, 133 (1997). One photon of wavelength λ = h/(m_ec) is confined to a closed path that goes twice round a torus. The paper assumes the confinement; it adds no field equation. From the field in a sphere of diameter λ/2 it states q = (1/2π)·√(3ε_0ħc) (eq. 5), the coupling α′ = q²/(4πε_0ħc) = (q/e)²α (eq. 8) and g = 2(1 + α′/2π) (eq. 12). Spin ½ comes from the double loop: the field returns to itself only after 720°.

Standard prediction. The Dirac electron with QED corrections, which gives the measured charge, magnetic moment and spin. Scattering shows the electron point-like below about 10^{-18} m. Under linear Maxwell equations nothing holds light in a loop, so it leaves at the speed of light.

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Part 1. The paper's numbers against measurement

quantity	model	measured (CODATA 2018; Fan et al. 2023)
charge, $q = (1/2\pi)\cdot\sqrt{(3\varepsilon_0\hbar c)}$ (eq. 5)	0.910318 e	1 e (exact)
coupling $\alpha' = (q/e)^2\cdot\alpha$ (eq. 8)	0.00604716	α = 0.00729735
magnetic anomaly $(g-2)/2 = \alpha'/2\pi$ (eq. 12)	0.000962436	0.00115965218
… with q set to e (= $\alpha/2\pi$, QED first order)	0.001161410	off by 0.0015
spin	1/2 (double loop)	1/2
size: Compton wavelength λ	2.426×10^{-12} m	point-like to ~10^{-18} m
loop radius $\lambda/4\pi$	1.931×10^{-13} m	

The model anomaly is 17.0 % below the measured one because the charge is 9.0 % below e.

Part 2. The double loop as a circuit: plain and crossed ring lines

A two-wire ring transmission line of 200 LC sections is an LC ladder closed on itself (a closed path in Kron's sense). It is built once plain and once with its two conductors swapped at one point. On the plain ring a wave returns to itself after one trip, so its modes hold a whole number of wavelengths. On the crossed ring the differential voltage changes sign at the crossing, so a wave matches itself only after two trips and the modes hold half-integer numbers of wavelengths. Both rings are solved in kronspice: an AC sweep finds the modes (the twin is the ladder's analytic dispersion) and a transient sends a 2.93 V pulse round. One trip takes 9.17 µs.

#	plain: AC peak	ladder theory	rel. diff	λ round ring	crossed: AC peak	ladder theory	rel. diff	λ round ring
1	109102 Hz	109104 Hz	-2.66×10^{-5}	1.0	54548.1 Hz	54553.9 Hz	-1.06×10^{-4}	0.5
2	218181 Hz	218182 Hz	-6.63×10^{-6}	2.0	163646 Hz	163648 Hz	-1.18×10^{-5}	1.5
3	327205 Hz	327206 Hz	-2.95×10^{-6}	3.0	272701 Hz	272702 Hz	-4.24×10^{-6}	2.5
4	436148 Hz	436149 Hz	-1.65×10^{-6}	4.0	381688 Hz	381689 Hz	-2.17×10^{-6}	3.5

pulse back at the start	trip 1	trip 2	trip 3	trip 4
plain ring	+2.65 V (+0.90)	+2.02 V (+0.69)	+1.88 V (+0.64)	+1.66 V (+0.57)
crossed ring	−2.68 V (−0.91)	+2.06 V (+0.70)	−1.72 V (−0.59)	+1.52 V (+0.52)

Fractions in brackets are of the 2.93 V launch. Ledgers: plain ring, in 3.438×10^{-8} J = heat 1.096×10^{-8} J + stored 2.341×10^{-8} J; crossed ring, in 3.438×10^{-8} J = heat 1.117×10^{-8} J + stored 2.32×10^{-8} J; time-stepping error 1.3×10^{-11} J in each.

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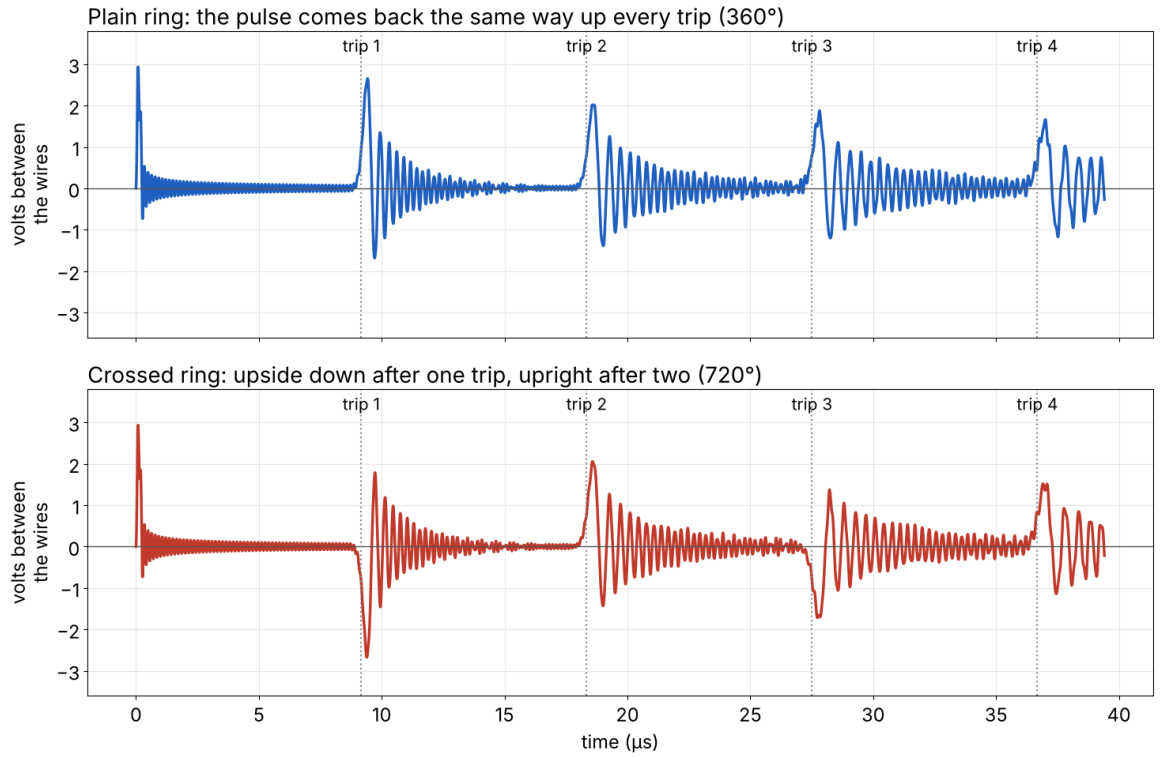


Figure 1. Differential voltage at the launch point. Top: the plain ring returns the pulse the same way up every trip (360°). Bottom: the crossed ring returns it inverted after one trip and upright after two (720°), the paper's spin-½ property in circuit form.

Part 3. Does light stay in the loop?

One wavelength of light is set on the paper's (2, 1) torus-knot path — twice round a ring of radius R = λ/4π while once round a tube of radius R/2 — as a travelling wave cos(ks − ωt) in a Gaussian tube of width R/3, with E and B across the path so the energy flows along it. E and B are built as discrete curls of sampled potentials, so the start is exactly divergence-free. Grid: λ = 1 m, R = 0.07958 m, cell 0.00663 m (R/12), region ±6R with an absorbing layer, path length 1.035 λ. The equations are linear, so lengths scale out. The solver is validated in *em/fdtd_test.go* (TM_{110} cavity to 10^{-7}, Hertzian dipole within 0.5 % of Larmor, flying doughnut within 3.8 %).

time (periods T)	0	0.125	0.25	0.5	1	1.5	2
energy within 2R	98.93 %	65.11 %	5.773 %	1.594×10^{-6} %	0.0001033 %	0.00173 %	5.549×10^{-7} %
in the region	100 %	100 %	100 %	90.42 %	0.3495 %	0.07582 %	0.01209 %
left the region	0 %	0 %	4.038×10^{-11} %	9.584 %	99.65 %	99.92 %	99.99 %

Ledger residual -1.9×10^{-13} of the starting energy; div B 7×10^{-13} (round-off).

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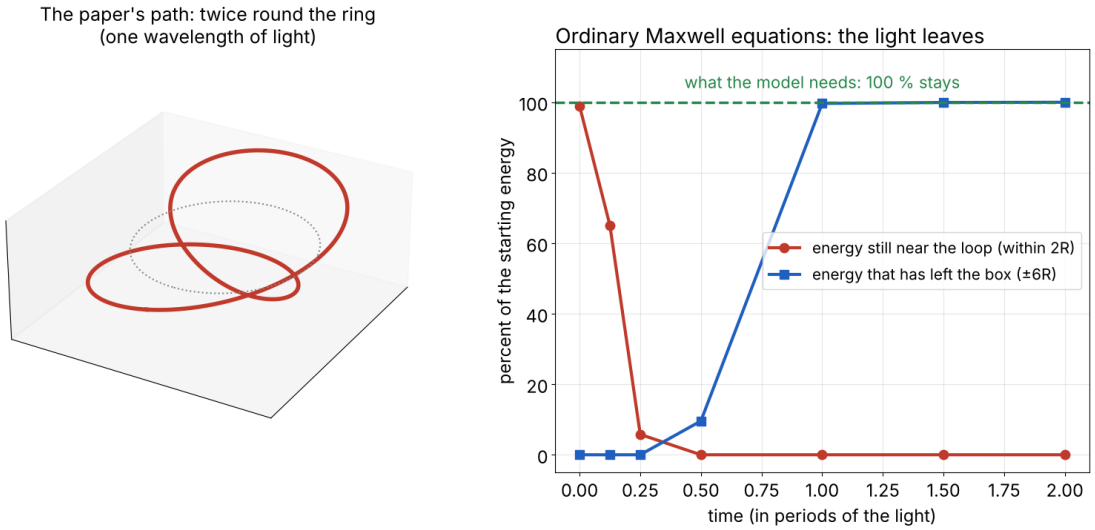


Figure 2. Left: the paper's path, twice round the ring (dotted: the ring's centre line). Right: share of the starting energy still within 2R of the loop (red) and that has left through the absorber at ±6R (blue); the dashed line is what the model needs, 100 % staying for ever.

Part 4. Three electron pumps that work

The model's own frequency, $m_e c^2/h = 1.24\times10^{20}$ Hz, is about 10^8 times beyond any circuit, so these pumps run at frequencies circuits reach. Electrons come from metal or ground; energy comes from a declared source.

4.1 Single-electron pump (Pothier et al. 1992)

Two aluminium islands between three tunnel junctions of 200 aF and 200 kΩ, two 50 aF gates driven 90° apart, at 20 mK; charging energy $e^2/2C = 0.178$ meV (2.1 K). Solved with the orthodox theory of single-electron tunnelling: a master equation over the islands' electron numbers with rates $\Gamma = -\Delta F/(e^2R(1 - e^{\Delta F/kT}))$, ΔF being the free-energy change of a tunnelling event.

drive	electrons per cycle	current	heat per electron (paid by the gates)	ledger
1 MHz	1.000000000	0.1602 pA	12.7 μeV	-4.1×10^{-10}
10 MHz	0.999999977	1.602 pA	53.1 μeV	-4.1×10^{-8}
30 MHz	0.999458493	4.804 pA	93.2 μeV	-3.7×10^{-7}
100 MHz	0.950531648	15.23 pA	160 μeV	-4.1×10^{-6}
300 MHz	0.667354712	32.08 pA	204 μeV	-1.0×10^{-5}

Errors grow when the drive outruns the tunnelling (RC ≈ 40 ps). Uphill against 20 μV at 10 MHz it still pumps 0.999998 electrons per cycle; the gates supply 72.8 μeV per cycle = 20.0 μeV lifted into the bias source + 52.8 μeV heat (ledger -5.6×10^{-8}).

4.2 Multipactor pump in a 1 GHz resonator (Farnsworth US 2,091,439; Kishek & Lau 1998)

A 1 mm, 1 cm² copper gap (as-received yield, Baglin 2002) is the capacitor of a resonator (C = 0.885 pF, L = 28.6 nH, Q = 1000) driven by 0.779 mA for 140 V unloaded; a 10 % grid passes electrons to a collector. The resonator alone agrees with a kronspice transient to 0.0031 of V_0. In step (140 V, 1000 seed electrons) the population grows to 1.98×10^8, pulls the gap voltage down to 39 V and stops multiplying; the resonator recovers and a stray electron (one per 20 ns while the gap is empty, declared; 7 used) restarts it, giving bursts as Kishek and Lau found. Out of step (300 V) no electrons are left after 40 periods. Over 2000 periods 2.16×10^8 electrons are collected, 15.2 μA on average, at 1426 eV of RF drive per collected electron.

Ledger [nJ]: drive 49.32 = resonator heat 22.98 + impacts on the plates 31.43 + collector 1.78 + stored change −6.876; residual -3.5×10^{-7}. The stored change is negative because the bursts also drain the energy the resonator held at the start.

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4.3 Ground charge pump with a resonant drive (§3j; Cockcroft & Walton 1932)

10 V at 100 kHz into a series resonator (Q = 20) → 3-stage negative ladder → 100 nF bank with a 10 MΩ load, 600 cycles, solved by kronspice. Step-up 10 V → 199.9 V peak (×20.0); bank −1194.7 V against the ideal −2·n·peak = −1199.5 V. The earth lends 7.46×10^{14} electrons to the bank's negative plate (plus 4.02×10^{12} through the load) and the bank's earthed plate returns the same number, so the earth's net charge is unchanged. Ledger: source 3.559 J = heat 3.425 J + stored 0.133 J (bank 0.0714 J) + time-stepping error 2.7×10^{-5} J.

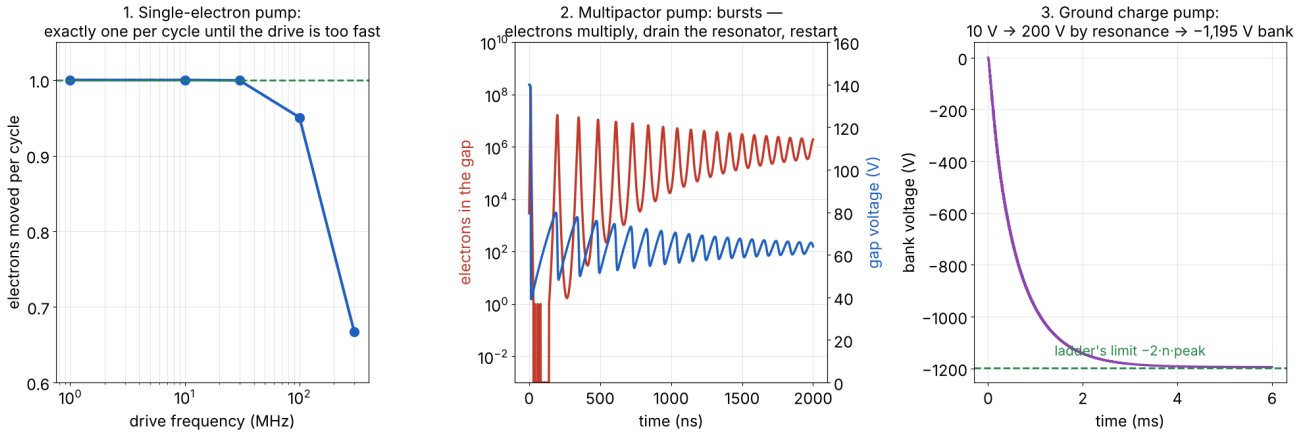


Figure 3. (1) Single-electron pump: electrons moved per cycle against drive frequency; exactly one until the drive is too fast. (2) Multipactor pump: electrons in the gap (red, log scale) and gap voltage (blue) — bursts that drain the resonator and restart. (3) Ground pump: bank voltage approaching the ladder's limit −2·n·peak.

Part 5. The multipactor pump on a microwave-oven magnetron, evolved

An LG 2M214 oven magnetron (2450 MHz, 900 W RF; efficiency 70 % declared, as the sheet gives it only as a chart; 34.6 W heater; 1320 W from the mains) feeds a circulator, a coupling port and a copper re-entrant cavity (Q_0 = 2000) whose capacitive gap multipacts, then a grid and a collector. Power the cavity does not take goes to the circulator's water load. The cavity's circuit twin is $C = \varepsilon_0 A/d$, L resonating it, wall loss $R_0 = Q_0/(\omega C)$, port $R_{ext} = R_0/\beta$ and a Norton source of the available power. The genetic algorithm (*evo*, value genome, seed 2450, 40 designs × 30 generations) chooses gap, gap area (side ≤ λ/10, so the gap stays lumped), coupling β, grid transparency and surface, for the fewest mains watts per collected microamp, below Kilpatrick's spark limit and 10 kV (non-relativistic). The search ran at ≥ 200 steps per period over 1500 periods; results were re-run at 800 steps over 2000 periods.

	best of generation 0	**evolved**
gap; area	2.761 mm; 0.376 cm²	1.519 mm; 1.500 cm²
coupling β; grid transparency	0.000139; 0.03	0.00153; 0.43
surface	copper, as received	copper, as received
gap amplitude, unloaded; peak field / Kilpatrick	1038 V; 0.009	1278 V; 0.020
collector current	**7.475 µA**	**659.4 µA**
mains W per µA; electrons per mains joule	176.6; 3.53×10^{10}	2.002; 3.12×10^{12}
into the cavity: wall heat + electrons	0.332 W: 0.297 + 0.0633	2.05 W: 1.79 + 0.816
to the water load	899.7 W	897.9 W
ledger residual	4.0×10^{-6}	3.9×10^{-6}

The collector current varies by about ±10 % with step size and run length (burst timing); the order of magnitude is stable. An earlier search found a false 0.3 A from time steps too coarse for fast electrons; the engine now takes at least 40 steps per gap crossing. The gap needs only 2.1 W, so one magnetron could feed about 438 such gaps (321 µA per watt into the cavity).

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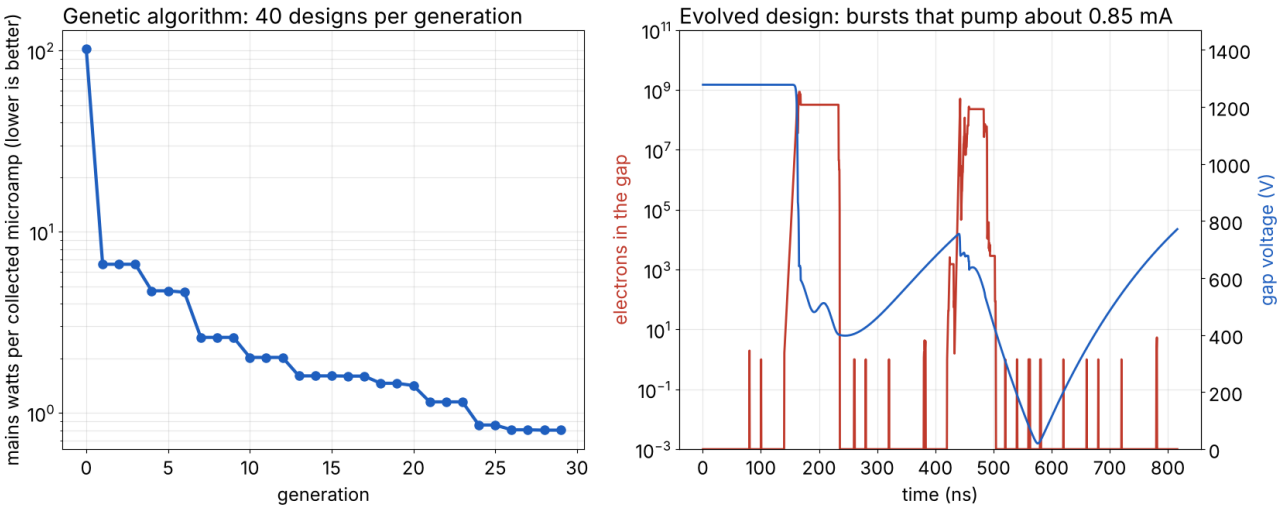


Figure 4. Left: best fitness (mains watts per collected µA) per generation, as measured during the search. Right: the evolved design's electrons in the gap (red) and gap voltage (blue) — bursts. The plotted search fitness (about 0.80 W/µA at the end) and the panel title (about 0.85 mA) are not the re-run values; the program's re-run at 800 steps per period gives 2.002 W/µA and 659.4 µA (table above).

Part 6. Scaling to amps: many gaps on one magnetron

The genetic algorithm also chooses how many identical gaps (1 to 1,000,000) share the magnetron through an ideal, lossless, isolated power divider (declared), each taking P_{out}/N.

	LG 2M214 (oven)	**Stellant CWM-100kW (industrial)**
frequency, RF power	2450 MHz, 1 kW	915 MHz, 100 kW
gaps in parallel	2686	786127
gap; area per gap	0.56 mm; 1.50 cm²	1.52 mm; 3.96 cm²
total gap area	0.403 m²	311 m²
gap voltage (unloaded)	191 V	196 V
collected current (per gap)	**1.95 A** (726 µA)	**216 A** (275 µA)
into the gaps / water load	0.695 / 0.205 kW	76.4 / 23.6 kW
mains	1.32 kW	119 kW
amps per kW of mains; per m² of gap	1.48; 4.84	1.83; 0.696
ledger residual (one gap)	6.7×10^{-7}	6.1×10^{-7}

Scaling the better design: 1 A needs about 0.548 kW from the mains and 1.44 m² of gap plates; 10 A about 5.48 kW and 14.4 m²; 100 A about 54.8 kW and 144 m²; 300 A about 164 kW and 431 m². The searches were still improving when they stopped, so these are lower bounds. Feeding thousands of gaps needs a real power-distribution structure (a large multi-gap cavity rather than a divider), which is not modelled. The electrons come from the copper; every watt comes from the magnetron.

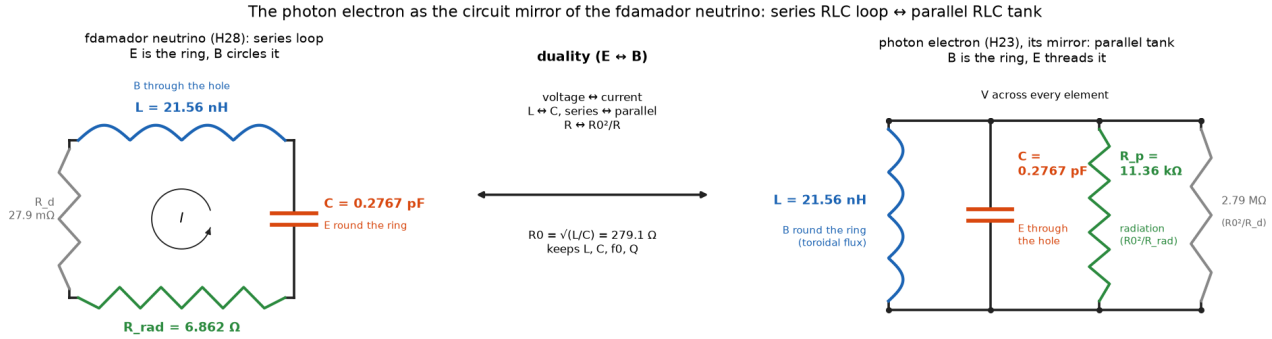
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The circuit mirror of the fdamador neutrino

The fdamador neutrino (H28, examples/fdamador_neutrino) is this example's field turned inside out: E is the ring and B circles it. In circuit form it is a series RLC loop: E round the ring is the capacitor, B through the hole the inductor, radiation a series resistor. Exchanging E and B is circuit duality (voltage and current, L and C, series and parallel exchange, and R becomes $R0^2/R$), so the photon electron's mirror is a **parallel RLC tank**: B round the ring (toroidal flux) is the inductor, E through the hole the capacitor, radiation a shunt resistor.

element	neutrino: series loop	electron mirror: parallel tank
inductor	L = 21.56 nH (B through the hole)	L = 21.56 nH (B round the ring)
capacitor	C = 0.2767 pF (E round the ring)	C = 0.2767 pF (E through the hole)
radiation	R_{rad} = 6.862 Ω in series	$R_p = R0^2/R_{rad}$ = 11.36 kΩ in shunt
dielectric loss	R_d = 27.9 mΩ in series	$R0^2/R_d$ = 2.79 MΩ in shunt
resonance, Q	f0 = 2.0606 GHz, Q = 40.51	f0 = 2.0606 GHz, Q = 40.51 (equal to 10^{-12})

The scale is R0 = √(L/C) = 279.1 Ω of the neutrino's bench loop (a = 10 mm, εr = 100; values from fdamador_neutrino's run_output.txt), which keeps L, C, f0 and Q. In the series loop every element carries the same current; in the tank every element sees the same voltage.



Both: f0 = 2.0606 GHz, Q = 40.51 (series ω0L/(R_rad+R_d) = parallel R_p‖R_pd/(ω0L), equal to 10^{-12}). In the series loop the current is shared, in the tank the voltage.
The mirror is exact as circuit algebra, for the models only. The real dielectric torus is not dual (εr = 100, μr = 1): its B-ring rings at 4.82 GHz, Q 112
(field solver, fdamador_neutrino Part 7b, r/2), not at the E-ring's 2.06 GHz. The H23 electron itself is a travelling wave on a ring line (E and B in phase, README §2) at 1.24×10²⁰ Hz;
a tank stands for one of its modes at a time. Under linear Maxwell its light leaves within ¼–½ period (README §3), Q of order 1. Neither circuit has mass, spin ½ from first principles or the weak interaction.

Figure 5. The fdamador neutrino's series RLC loop, the duality, and the photon electron's parallel RLC mirror, with values (viz/plot_circuit.py). An animation of one cycle, the light travelling along the (2, 1) path beside the mirror tank's energy moving between C and L, is viz/cycle.gif.

What the mirror is and is not. It is exact circuit algebra between the two **models**, not a statement about the measured electron or neutrino. The real dielectric torus is not dual (εr = 100, μr = 1): its B-ring mode rings at 4.82 GHz with Q 112 in the field solver (fdamador_neutrino Part 7b, grid r/2), not at the E-ring's 2.06 GHz. The H23 electron itself is a travelling wave on a ring line, with E and B rising and falling together (Part 2), whereas in the tank they alternate; a lumped tank stands for one of its modes at a time, and under linear Maxwell its light leaves within a quarter to half a period (Part 3). Neither circuit has mass or the weak interaction, and neither derives spin ½.

Ledger and validation

test	checks
TestPaperNumbers	eqs. 5, 8 and 12 of the paper
TestRingModes, TestRoundTrips	whole- and half-wave modes against the ladder; upright and inverted returns; ledgers close
TestKnotLightLeaves	on a coarse grid the light leaves, the field stays divergence-free, the ledger closes
TestSingleElectronPump	one electron per cycle at low drive, uphill too, the gates paying
TestMultipactorPump	resonator matches kronspice; growth in step, death out of step; ledger
TestGroundPump	the bank reaches −2·n·peak to 1 %; ledger
TestMagnetronEvolution	cavity matches kronspice; a short evolution improves on its start inside the limits
TestGapArray	N gaps are N copies of one gap fed P_{out}/N; the 100 kW tube's declared numbers

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Limits of the model

- Part 3 follows one initial field configuration under linear Maxwell equations; it shows the configuration is not self-confined, not that no confining mechanism could exist.
- The ring lines reproduce the 720° topology of the double loop, not the photon or the electron.
- The magnetron efficiency (70 %) and the lossless power divider are declared; the divider stands in for a multi-gap cavity that is not modelled. Collector currents vary by about ±10 % with step size.
- The stray-electron rate of the multipactor pump (one per 20 ns while the gap is empty) is declared.

Status

H23 is **bounded**. The model charge is 0.910 e and its anomaly 17 % low against Fan et al. (2023); spin ½ from the double loop is reproduced as the crossed ring's half-integer modes and its sign flip per trip; the self-confinement is assumed, not derived, and under linear Maxwell the configuration leaves in less than half a period. The model needs a confining mechanism it does not give. Standard counterpart: the Dirac electron with QED.

Reproducing the results

```
go run ./examples/photon_electron -viz examples/photon_electron/viz   # about 10 minutes
python3 examples/photon_electron/viz/render.py                         # pictures
go test ./examples/photon_electron                                    # about 1 minute
KRONSPICE_EXPORT=1 go test ./examples/photon_electron -run TestExport  # netlists, results sheets
python examples/photon_electron/viz/make_report.py examples/photon_electron/photon_electron_report.pdf
```

Sources: Williamson & van der Mark, Ann. Fond. Louis de Broglie 22, 133 (1997); Fan et al., PRL 130, 071801 (2023); CODATA 2018; Yee (1966); Hellwarth & Nouchi, PRE 54, 889 (1996); Pothier et al., Europhys. Lett. 17, 249 (1992); Grabert & Devoret (1992) ch. 2; Farnsworth US 2,091,439; Kishek & Lau, PRL 80, 193 (1998); Baglin et al., LHC Project Report 472 (2002); Kilpatrick, Rev. Sci. Instrum. 28, 824 (1957); Cockcroft & Walton, Proc. R. Soc. A 136, 619 (1932); LG 2M214 and Stellant CWM-100kW sheets.

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